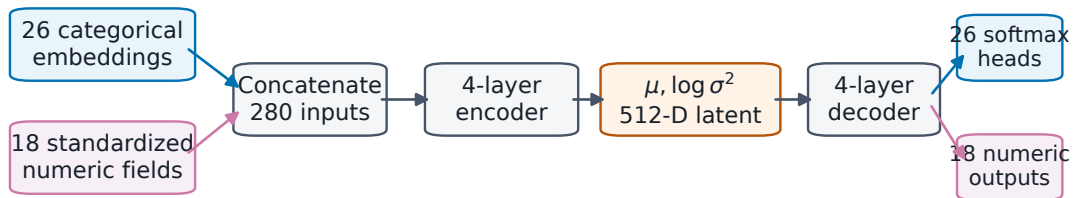
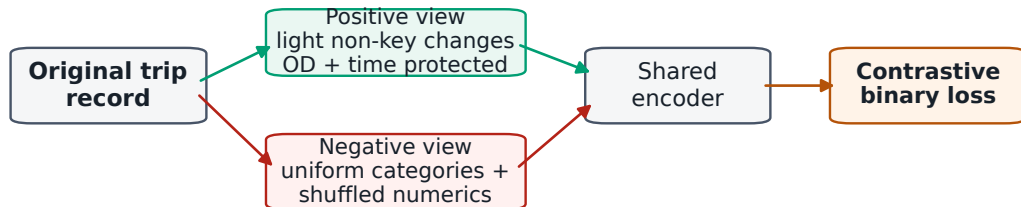


(A) Mixed-tabular variational autoencoder



Survey expansion weights are capped at the 99th percentile and mean-normalized for training; weight is never generated.

(B) Trip-specific contrastive regularization



Training objective: reconstruction + 0.03 KL + 0.25 L_{con} ; the ablation sets the final coefficient to zero.